

India's Agricultural Crop Production Analysis

TEAM ID	SWTID-2026-4021
PROJECT TITLE	India's Agricultural Crop Production Analysis
MAX MARKS	5 MARKS

CHAPTER – VII

FUNCTIONAL AND PERFORMANCE TESTING

7.1 – Performance Testing & Functional Testing

Here is a step-by-step structured guide you can use for a Tableau project on “India’s Agricultural Crop Production Analysis”, including data handling, filters, calculations, and visualizations.

India’s Agricultural Crop Production Analysis (Tableau Project Guide)

Step 1: Define Objective

The main goal is to analyze:

- Crop production trends in India
- State-wise contribution
- Crop-wise comparison (Rice, Wheat, Pulses, etc.)
- Year-wise production growth
- Regional agricultural performance

Step 2: Data Preparation (Amount of Data Rendered to Tableau)

Dataset Structure (Typical Fields)

You should prepare a dataset like:

Column Name	Description
Year	Agricultural year
State	Indian state name
Crop_Type	Rice, Wheat, Pulses, etc.

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Column Name	Description
Season	Kharif / Rabi / Zaid
Area_Ha	Area under cultivation
Production_Tonnes	Total production
Yield	Production per hectare
Rainfall	Optional
Irrigation_Type	Optional

Amount of Data Rendered to Tableau

- Rows: 50,000 – 2,00,000 records (ideal for meaningful dashboard)
- Time period: 10–20 years (e.g., 2005–2025)
- Geography: 28 states + UTs
- Crops: 8–12 major crops

Example:

- 20 years × 28 states × 10 crops ≈ 56,000 rows

Step 3: Data Import into Tableau

1. Open Tableau Desktop
2. Click Connect → Excel / CSV / Database
3. Load dataset
4. Check data types:
 - Year → Date/Integer
 - Production → Measure (Number)
 - State → Dimension

Step 4: Data Cleaning in Tableau

- Remove null values
- Standardize state names (e.g., “A.P.” → “Andhra Pradesh”)
- Convert production units if needed (Tonnes/Lakh Tonnes)
- Ensure correct data type assignment

Step 5: Utilization of Filters in Tableau

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Filters help users interact with data.

Key Filters Used:

1. Year Filter
 - Range slider (e.g., 2010–2025)
2. State Filter
 - Multi-select dropdown
3. Crop Type Filter
 - Rice, Wheat, Pulses, Oilseeds
4. Season Filter
 - Kharif / Rabi / Zaid
5. Production Range Filter
 - High / Medium / Low production
6. Top N Filter
 - Top 10 producing states

Purpose:

- Focus analysis
- Compare regions
- Track trends over time

Step 6: Number of Calculation Fields (Tableau Calculations)

You can create 5–8 calculated fields.

Important Calculation Fields:

1. Yield Calculation

Yield = Production / Area

2. Growth Rate (%)

((Current Year Production - Previous Year Production) / Previous Year Production) * 100

3. Rank of State

RANK(SUM([Production]))

4. Total Production per State

SUM([Production])

5. Crop Contribution %

(SUM(Production) / TOTAL(SUM(Production))) * 100

6. Moving Average (Trend)

WINDOW_AVG(SUM([Production]), -2, 0)

Recommended:

Minimum: 5

calculatedfields

Advanced project: 8–10 fields

Step 7: Number of Visualizations / Graphs

You should build 6–10 dashboards/graphs.

Core Visualizations:

1. Line Chart – Production Trend

- X-axis: Year
- Y-axis: Total Production
- Shows overall agricultural growth

2. Bar Chart – State-wise Production

- Compare top states like UP, Punjab, Maharashtra, AP

3. Pie Chart – Crop Share

- Rice, Wheat, Pulses, Oilseeds contribution

4. Map Chart – Geographic Distribution

- India map showing production intensity by state

5. Area Chart – Crop-wise Trend Over Time

- Shows how each crop changes over years

6. Heatmap – State vs Crop Performance

- Dark color = high production

7. Treemap – Crop Contribution

- Visual hierarchy of crop production share

8. Scatter Plot – Yield vs Area

- Shows efficiency of farming

Step 8: Dashboard Creation

Combine all charts into 2–3 dashboards:

Dashboard 1: Overview

- Total production
- Yearly trend
- Top states

Dashboard 2: Crop Analysis

- Crop share
- Crop-wise trends
- Yield comparison

Dashboard 3: Geographic Insights

- Map view
- Heatmap
- State comparison

Step 9: Final Insights

You can conclude:

- Rice and wheat dominate production
- Punjab, UP, and Maharashtra are top contributors
- Yield varies significantly by state
- Irrigation improves productivity
- Climate variability affects output

Summary (For Report Writing)

India's Agricultural Crop Production Analysis

Component	Value
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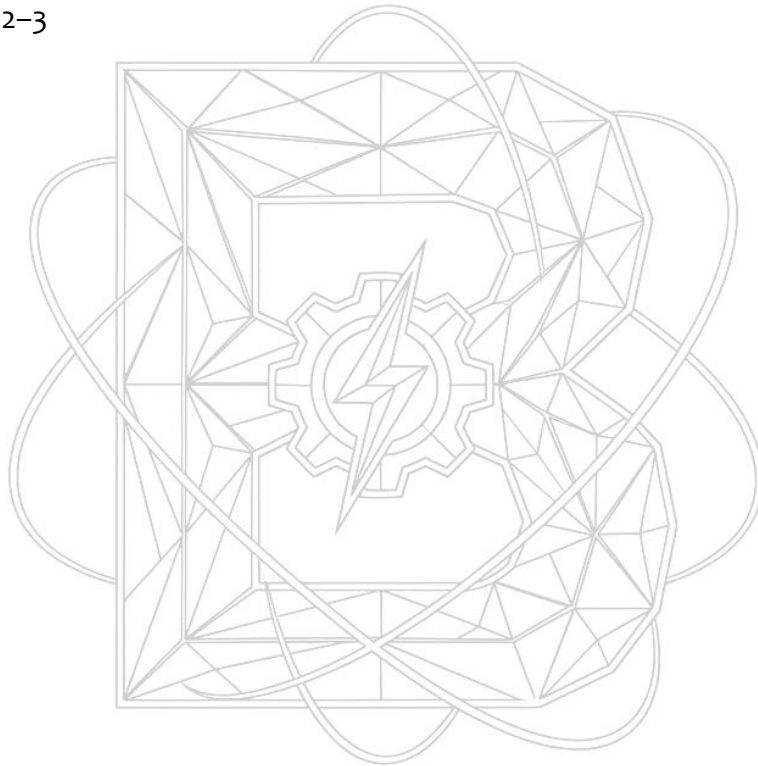
Data Size	50K–200K rows
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Filters Used	4–6
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Calculated Fields	5–8
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Visualizations	6–10
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Dashboards	2–3
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BRAVO
— BUILDERS —
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